

CHHEANG UDOMVEASNA

Data Analyst Student

EDUCATION

Department of Mechanical Engineering

Institute of Technology of Cambodia

- GPA 3.0/4.0 (1st Year)
- GPA 2.0/4.0 (2nd Year)

2023 - 2024

Department of Applied Mathematics and Statistics

Institute of Technology of Cambodia

- GPA 3.5/4.0 (3rd Year)
- GPA 3.5/4.0 (4th Year)

2024 - Present

General English Program

Australian Centre for Education(ACE)

2025 - Present

General English Program

Camfirst School

2016 - 2018

CONTACT

Phone : 096 671 9476

Email : udomveasnachheang@gmail.com

Address : Phnom Penh, Cambodia

LinkedIn : Udomveasna Chheang

PROFILE

I am a Data Analyst student at the Institute of Technology of Cambodia. I have experience in data analysis, SQL, Power BI, Excel, and Python, with a strong foundation in data visualization, statistical analysis, and database management. Through academic and personal projects, I have developed skills in transforming raw data into meaningful insights to support decision-making. I am seeking a Data Analyst Internship where I can apply my analytical skills, gain hands-on experience, and continue growing in the field of data analytics.

EXPERIENCES

ITC Volunteer — **2022 – 2023**

- Provided logistical support
- Organize the openhouse events
- Guided Attendees to tour all laboratories

Sea Game Volunteer — **2023**

Provided guidance on hotel locations, transportation, and event venues to ensure smooth arrivals and check-ins. Acted as a point of contact for visitors by offering directions, answering questions, and coordinating with other volunteers to deliver efficient hospitality services throughout the event.

Dc Lab Robotics — **2023 – 2024**

- Engaged in pre-intermediate research and hands on project focused on robotics systems design, coding, and integration.
- Collaborate on building and programming autonomous robots, blending mechanical engineering with control systems.

SKILLS & TECH SKILLS

- Management Skill
- Data Analytic
- Create Dashboard
- Data Visualization
- Teamwork
- Communication Skills
- Time Management
- Database
- Microsoft Tools

LANGUAGE

- Khmer (Native)
- English (Upper-Intermediate)
- French (Basic)

REFERENCES

PHAUK Sokkhey, Ph.D

Deputy Head of Department of Applied Mathematics and Statistics (AMS) & Head Master Program of Data Science.

Phone : 012 939 310

Email : phauk.sokkhey@itc.edu.kh

LONG Pakrigna, Mr.

Researcher and Lecturer, Department of Applied Mathematics and Statistics (AMS)

Phone : 010 644 496

Email : long.pakrigna@itc.edu.kh

KEY PROJECTS

Bilingual Chatbot (Khmer - English) **2024 – 2025**

- Developed a bilingual (Khmer-English) AI chatbot capable of answering educational questions using Retrieval-Augmented Generation (RAG).
- Collected and processed Grade 12 Khmer textbook datasets to create a knowledge base for subject-specific question answering.
- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline, including document collection, text preprocessing, chunking, embedding generation, and semantic retrieval.
- Integrated GPT-4o with a vector database to generate context-aware responses grounded in textbook content, improving answer accuracy and reducing hallucinations.
- Technologies: Python, GPT-4o, LangChain, FAISS/ChromaDB, NLP, RAG.

Intelligent Face Attendance System Using YOLO11 & ArcFace (Applied in IES Conference in indonesia) **2025 – 2026**

- Developed an AI-powered attendance system using YOLO11 face detection and ArcFace facial recognition for real-time student identification.
- Built a schedule-aware attendance validation system integrated with PostgreSQL to automate attendance tracking and prevent duplicate records.
- Designed a Streamlit dashboard for attendance analytics, reporting, and administrative monitoring.
- Achieved 100% face detection rate during evaluation and deployed a complete end-to-end computer vision pipeline.
- Technologies: Python, YOLO11, ArcFace, OpenCV, PostgreSQL, Streamlit, ONNX Runtime.

Smart Air-Conditioning Control System (IoT) (Applied in IES Conference in indonesia) **2025 – 2026**

- Developed an IoT-based smart AC control system using ESP32, Firebase, and environmental sensors for automated temperature and occupancy-aware control.
- Implemented a rule-based automation engine with conflict detection and cloud monitoring capabilities.
- Built a real-time dashboard for monitoring sensor data, system status, and remote management.
- Achieved 100% automation rule accuracy during system validation.
- Technologies: ESP32, Arduino, Firebase, REST API, IoT, Embedded Systems, HTML/CSS/JavaScript.